

Database Systems

Assignment 02 – Fall 2024

Instructions:

1. Please submit a pdf file on LMS with the name format: **DB-A02-YOUR_REGISTRATION_NO-Fall2024.pdf** (e.g. DB-A02-Num1-F23-12345).
 2. Note: Only the placeholders in red will be replaced with your respective ones.
 3. Make sure your complete names and roll numbers are clearly mentioned on the title page of the deliverable that you submit.
 4. If your submission is late or doesn't follow the naming scheme/instructions, it will not be graded.
 5. **Due Date: 11:59 pm - October 22, 2024.**
 6. This is an individual hand-written assignment. Submit a scanned copy.
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Task:

Based on your understanding of Entity-Relationship modeling, analyze the given case studies and provide their ERDs. The data requirements and the query functions are given for all the case studies. You are required to provide the entities with their attributes, the relationships between entities along with the cardinality constraints.

XYZ Hospital wants to develop a Patient Management System to efficiently handle patient records, doctor consultations, and hospital resources.

- Patients have personal information such as a unique ID, name, contact details, and insurance provider. A patient can make appointments with doctors for consultations.
- Doctors specialize in one or more departments (e.g., cardiology, neurology). Each doctor has a unique ID, name, specialization, and contact details. Doctors can consult multiple patients and prescribe treatments.
- The hospital is divided into several departments, such as emergency, surgery, and outpatient care. Each department has a name, head of department, and list of doctors.
- Appointments are made by patients to consult doctors. Each appointment has a unique ID, date, time, and consultation notes. A patient can have multiple appointments with the same doctor or different doctors.
- Treatments can be prescribed to patients by doctors. Each treatment includes medication or procedures that are stored in the hospital inventory. The inventory tracks the quantity and supplier of each item.
- Patients may be admitted for in-patient care, where they are assigned to a ward. Each ward has a unique ID, location, and capacity, and is supervised by a department.

Case Study 2

Global Delivery Services (GDS) is an international logistics company that specializes in delivering parcels and freight across various countries. The company has a vast network of warehouses, distribution centers, and offices worldwide. GDS is looking to implement a comprehensive database system to streamline its operations, manage client information, and track delivery processes more efficiently.

GDS serves both corporate and individual clients. Each client has a unique client ID, name, address, and contact details. Corporate clients also have a tax identification number and may have multiple branches, each represented with a unique branch ID, address, and manager details. Individual clients often use the service for personal deliveries, whereas corporate clients frequently send large shipments, requiring additional logistics coordination. Clients can request shipments that involve transporting parcels or goods between different locations. Each shipment has a unique shipment ID, details about the contents, dimensions, and weight, as well as pickup and delivery addresses. The system also tracks the pickup date and expected delivery date. For corporate clients, shipments can sometimes involve sending goods to multiple locations in a single request, creating multiple delivery orders, each with its own unique order ID.

Each shipment is assigned to a warehouse or distribution center, depending on its size and destination. Warehouses are located in different countries, with each facility assigned a unique warehouse ID, address, capacity, and manager. A warehouse can store multiple shipments until they are dispatched for delivery, and each warehouse has a warehouse manager responsible for overseeing the inventory. The database must track how long a shipment stays in a warehouse, as well as the space it occupies.

Shipments are transported through various carriers, which can be GDS's own fleet or third-party logistics companies. Each carrier has a unique carrier ID, name, and contact information. The system needs to record which carrier is responsible for each shipment, and in the case of large shipments, multiple carriers might handle different legs of the journey (e.g., air, sea, and road). The system also tracks carrier schedules, which include departure and arrival times, the route taken, and the carrier's assigned vehicle or aircraft. Each carrier has a list of vehicles or transport methods, each with a unique vehicle ID, vehicle type (e.g., truck, plane), capacity, and registration number.

To ensure timely deliveries, GDS offers tracking services for its shipments. Each shipment is assigned a tracking number, which clients can use to check the status of their delivery. The system must record each tracking event, such as when a shipment is picked up, arrives at a warehouse, departs a warehouse, and is delivered to the final destination. The status of each shipment (e.g., in transit, delayed, delivered) must be updated in real-time, and reasons for delays (e.g., customs clearance, weather) must be recorded. Additionally, GDS provides insurance options for shipments. Each shipment can be insured, and the insurance record must include the insurance policy number, insured amount, and the name of the insurance provider.

Furthermore, the system must handle employee information. GDS employs staff at various levels, including delivery personnel, warehouse workers, customer service agents, and logistics managers. Each employee is assigned an employee ID, name, position, and department. Delivery personnel are responsible for ensuring that shipments reach their destination on time and can be assigned to multiple deliveries in a single day. Employees are also associated with specific offices or warehouses, with each office having a unique office ID and location details. Customer service agents handle client inquiries, including requests for pickup and complaints about delays or damaged goods.

In addition, GDS wants to integrate a feedback system where clients can rate their delivery experience. Feedback includes a rating (out of five stars) and optional comments. Each feedback entry is tied to a specific shipment, and clients can only leave feedback once their shipment has been successfully delivered. The company also wants to track client satisfaction over time, linking it to delivery performance metrics, such as on-time delivery percentage and the number of delayed or damaged shipments.

Finally, GDS is looking to implement a billing system. The system must generate invoices for both individual and corporate clients. Each invoice is associated with one or more shipments and includes details such as the total amount due, payment status, and payment date. Invoices are linked to specific payment methods, which can include credit card, bank transfer, or corporate account. In the case of corporate clients, payments may be aggregated across multiple shipments, and discounts can be applied based on the client's contract terms or shipping volume. The system should also track late payments and apply penalties when necessary.